

Why I Chose Phi-3 Mini 4K Instruct Over Alternatives

I selected the Microsoft Phi-3 Mini 4K Instruct model for this project because it provides the best balance between performance, efficiency, and structured output generation for an end-to-end conversational analysis pipeline.

Compared to larger models such as Mistral-7B, Phi-3 Mini is lightweight and can run efficiently on Google Colab without requiring high-end GPU resources. This made development and testing faster and more practical.

Compared to smaller alternatives like Gemma 2B, Phi-3 Mini provides better instruction-following capability and generates more reliable structured JSON responses, which was important for generating:

- call summaries,
- sentiment analysis,
- agent scores,
- strengths,
- improvement areas,
- and recommended next steps.

Another important reason for choosing Phi-3 Mini is that it is fully open-source and aligned with the assignment requirement of using open-source Small Language Models only.

Overall, Phi-3 Mini provided:

- efficient inference,
- strong instruction following,
- clean structured outputs,
- and easy deployment in Google Colab.

Simple Architecture Diagram

